# Lopcombe Corner — Rainfall vs Peak Level (Oct–Dec & antecedent minimum)

25 winters 2000/01–2024/25 · levels: EA logger E14150 (winter peak = max Oct–Apr) · rainfall: Broughton E13580 daily totals summed (±few %) · analysis 7 Sep 2026

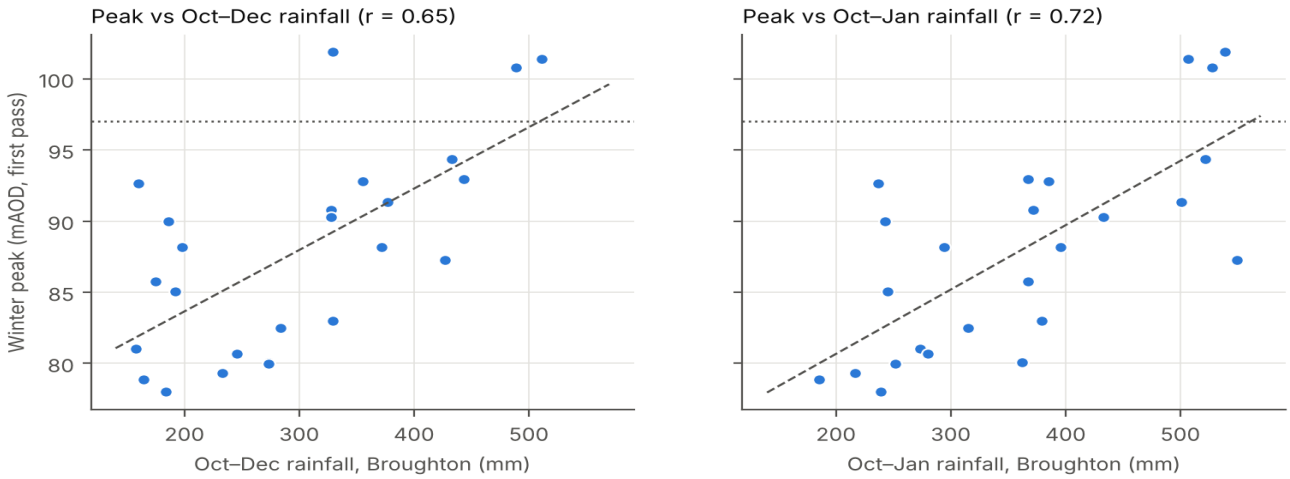
**Recreated 20 Oct 2026 — superseded.** These results used the first-pass winter peaks, three of which were later found to be truncated (2019/20, 2020/21, 2023/24, each ~4.5 m too low), and Broughton totals that under-read in wet winters. See the Oct–Mar report and the HadUK-Grid analysis of 20 Oct 2026 for the current figures. The conclusion here — keep a window that includes January or later — still holds.

## Headline

- **Oct–Dec does not beat Oct–Jan.** Peak vs Oct–Jan r = 0.72 ($r^2$ 0.52); vs Oct–Dec r = 0.65 ($r^2$ 0.43); both p < 0.001. Most winter peaks occur Feb–Apr, so January rain genuinely feeds them. The Feb-2014 record grew from a Dec 2013 / Jan 2014 deluge, and Oct–Dec alone under-predicts it by about 13 m.
- **Antecedent recession minimum helps only slightly here.** Adding it to Oct–Dec lifts $R^2$ from 0.43 to 0.49 (partial r ≈ 0.32, p > 0.05, n = 25). The direction supports the buffer idea — 2009 peaked at 92.6 m on only 160 mm Oct–Dec from the highest autumn start (79.4 m) — but the well recedes to a near-constant autumn baseline (78.5 ± 0.4 m), so it carries little multi-year memory.

## Model comparison

| Model | r / $R^2$ | adj $R^2$ | Note |
|---|---|---|---|
| peak ~ Oct–Jan rain | r 0.72 / 0.52 | 0.50 | best single predictor |
| peak ~ Oct–Dec rain | r 0.65 / 0.43 | 0.40 | weaker |
| peak ~ Oct–Dec + antecedent min | 0.49 | 0.44 | antecedent not significant |
| peak ~ Oct–Jan + antecedent min | 0.56 | 0.52 | best overall |




## 97 m Pitton threshold

Breaches in this dataset (2001, 2003, 2014) all needed a very wet winter, but the Oct–Dec total is a poor trigger: 2014 breached on only 329 mm Oct–Dec, while 2024, 2013 and 2021 had more Oct–Dec rain (443/433/427 mm) and did not breach. Timing within the winter and antecedent state gate the threshold, so a single autumn rainfall total should not be used as the flood trigger.

## Data (as used in September 2026)

| Winter ending | Oct–Dec mm | Oct–Jan mm | Peak mAOD | Antecedent min mAOD |
|---|---|---|---|---|
| 2001 | 511 | 507 | 101.39 | 78.65 |
| 2002 | 192 | 245 | 85.05 | 78.55 |
| 2003 | 489 | 528 | 100.79 | 78.71 |
| 2004 | 284 | 315 | 82.48 | 78.30 |
| 2005 | 273 | 362 | 80.05 | 78.10 |
| 2006 | 233 | 217 | 79.33 | 77.70 |
| 2007 | 328 | 372 | 90.76 | 77.58 |
| 2008 | 175 | 367 | 85.76 | 78.92 |
| 2009 | 160 | 237 | 92.62 | 79.36 |
| 2010 | 372 | 396 | 88.15 | 78.17 |
| 2011 | 158 | 273 | 81.05 | 78.25 |
| 2012 | 184 | 239 | 77.99 | 77.99 |
| 2013 | 433 | 522 | 94.35 | 79.02 |
| 2014 | 329 | 539 | 101.89 | 78.58 |
| 2015 | 329 | 379 | 83.00 | 78.86 |
| 2016 | 198 | 294 | 88.15 | 78.56 |
| 2017 | 164 | 185 | 78.88 | 78.70 |
| 2018 | 186 | 243 | 89.98 | 78.47 |
| 2019 | 246 | 280 | 80.66 | 78.40 |
| 2020 | 355 | 385 | 92.81 | 77.93 |
| 2021 | 427 | 549 | 87.24 | 78.66 |
| 2022 | 273 | 252 | 79.96 | 78.67 |
| 2023 | 377 | 501 | 91.35 | 78.07 |
| 2024 | 443 | 367 | 92.97 | 78.92 |
| 2025 | 328 | 433 | 90.28 | 79.08 |

Known issues found later: in 2001, 2006, 2022 and 2024 the Oct–Jan sum is smaller than Oct–Dec, which is impossible and shows the fetch dropped days; the 2020, 2021 and 2024 peaks are truncated (corrected 97.25, 91.72 and 97.54 m).